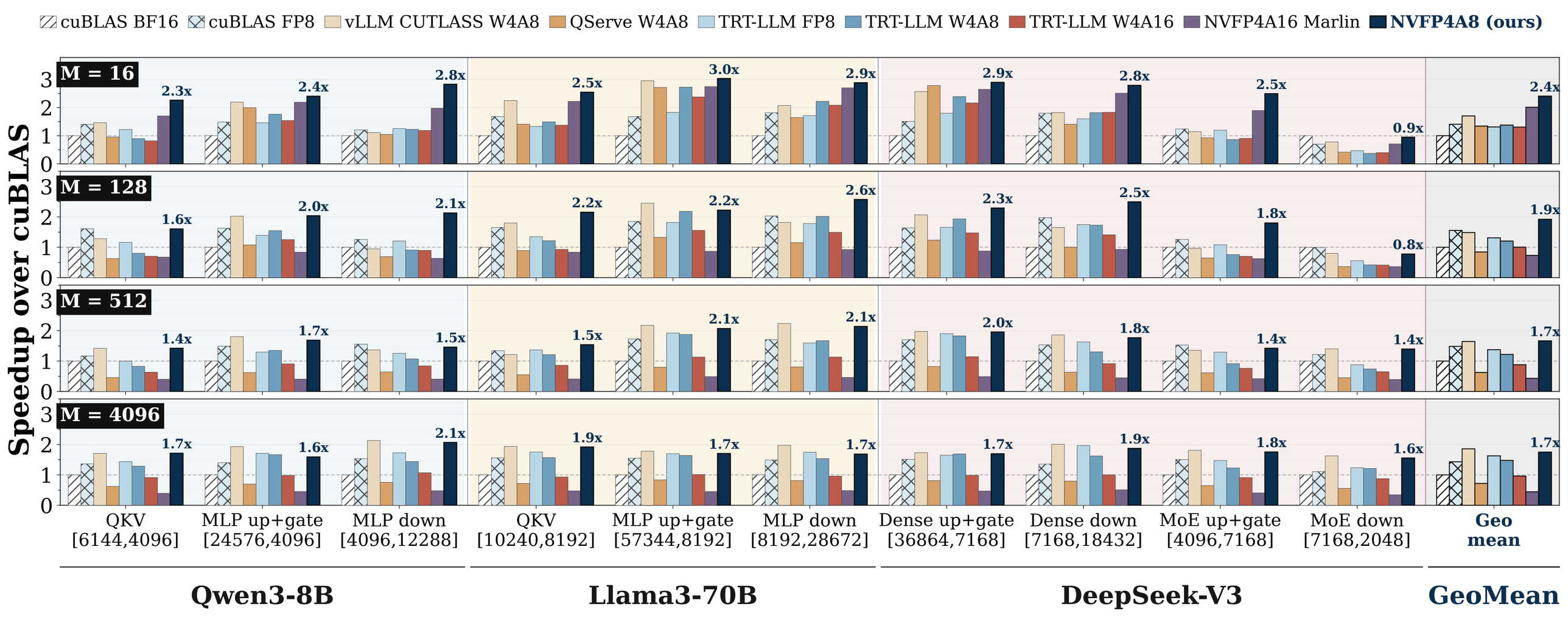
cuBLAS BF16
cuBLAS FP8
vLLM CUTLASS W4A8
QServe W4A8
TRT-LLM FP8
TRT-LLM W4A8
TRT-LLM W4A16
NVFP4A16 Marlin
NVFP4A8 (ours)
Speedup over cuBLAS
M = 16
M = 128
M = 512
M = 4096
2.3x
2.4x
2.8x
2.5x
3.0x
2.9x
2.9x
2.8x
2.5x
0.9x
2.4x
1.6x
2.0x
2.1x
2.2x
2.2x
2.6x
2.3x
2.5x
1.8x
0.8x
1.9x
1.4x
1.7x
1.5x
1.5x
2.1x
2.1x
2.0x
1.8x
1.4x
1.4x
1.7x
1.7x
1.9x
2.1x
1.9x
1.7x
1.7x
1.7x
1.9x
1.8x
1.6x
1.7x
QKV [6144,4096]
MLP up+gate [24576,4096]
MLP down [4096,12288]
QKV [10240,8192]
MLP up+gate [57344,8192]
MLP down [8192,28672]
Dense up+gate [36864,7168]
Dense down [7168,18432]
MoE up+gate [4096,7168]
MoE down [7168,2048]
Geo mean
Qwen3-8B
Llama3-70B
DeepSeek-V3
GeoMean